HACKATHON 2 · PUNJAB JOBS AI BOOTCAMP · J-PAL SOUTH ASIA

Personalised Offer Engine

Decides which offer each customer should get (or none), lets an AI personalise it inside business rules, checks every offer, and delivers it on Telegram.

Try it: t.me/oooHusanBot

Code: github.com/Hussnn18/h2

Husanpreet · October 2026 · all customer data is synthetic

How it’s built

1 · DATA

Synthetic shoppers

500 Punjab customers; a hidden type (deal hunter, loyal…) drives behaviour

2 · RULES

Rules set the limits

Which offers each customer may get, and the range: caps, budget, frequency

3 · AI

Claude personalises

Picks one allowed offer and value, writes the message, or sends nothing

4 · CHECK

Validator checks all

Re-checks every limit; the AI never writes numbers; else the rules’ pick

5 · DELIVER

Telegram bot

Offer with code and expiry; buttons Claim, Not for me, Why this offer?

6 · MANAGE

Marketing portal

Edit offers and rules, preview any customer, see results; all logged

Stack: Python · python-telegram-bot · Claude API (structured output) · SQLite · Streamlit · 43 automated tests

How to test it

THE LIVE BOT

t.me/oooHusanBot

1. Press **Start**: you play a synthetic customer
2. Read your profile card, then your first offer
3. Tap **Why this offer?**, then Claim or Not for me
4. Send /offer again at once: the gap rule says no
5. /newpersona for another customer, /stop to opt out

PORTAL, CODE AND DATA

Marketing portal

- Code: github.com/Hussnn18/h2
- Run locally: python -m streamlit run portal/app.py
- Test data: data/sample/*.csv (500 customers, 2,238 orders)
- Or regenerate: python -m data.generate
- Tests: python -m pytest (43 pass)

FEATURE 1

Personal offers on Telegram

Each classmate plays a synthetic customer. The bot sends the offer that earns the most **extra** profit for that customer, or nothing.

1. Open t.me/oooHusanBot, press Start
2. See your profile card and first offer
3. Code, minimum order and expiry are added by code, never by the AI
4. Tap Why this offer? for the reason and the rules it passed

[Add phone screenshot: an offer with its Claim / Not for me / Why buttons]

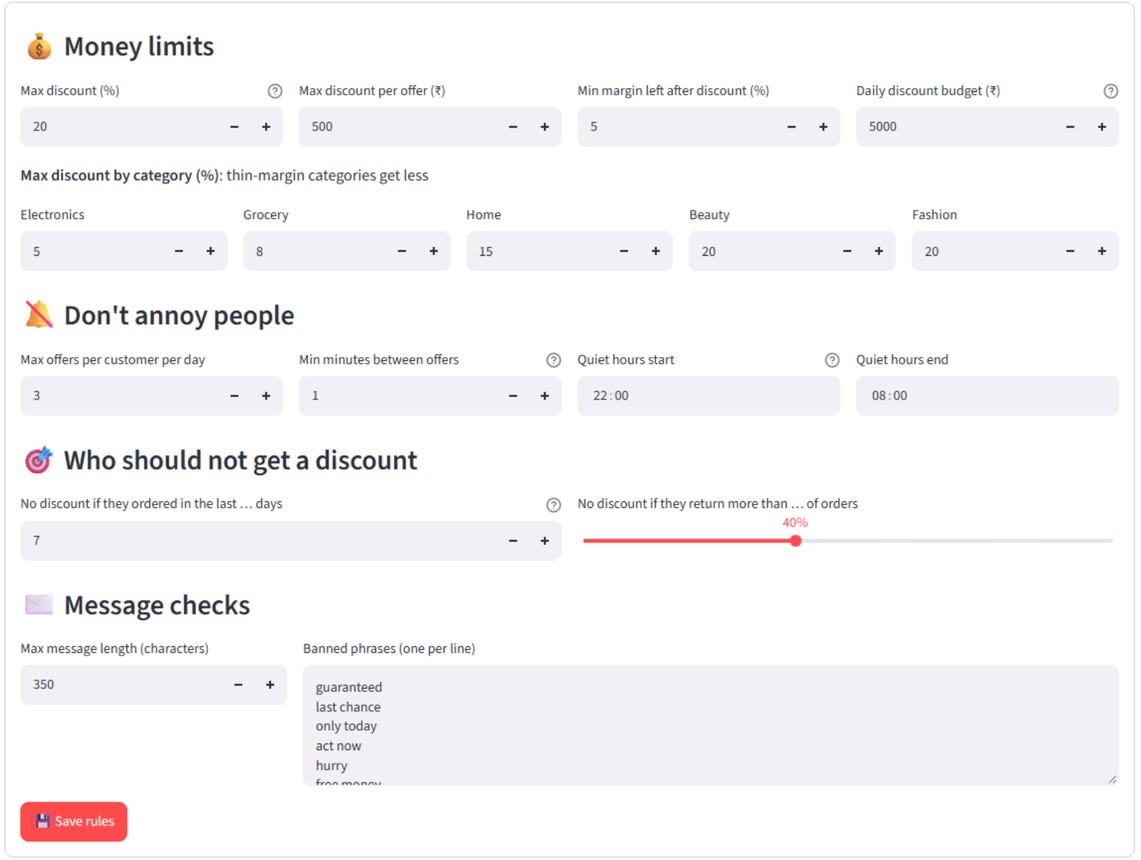
FEATURE 2

Business rules the AI can’t break

Limits every offer must respect, whoever picked it. Marketing changes them without code.

1. Portal → Rules
2. Change a limit: max %, ₹ cap, category caps, budget, offers per day, quiet hours
3. Save rules: checked, then used from the next offer

Tested: an AI answer of “90% off, last chance!” is blocked and the rules’ pick is sent.

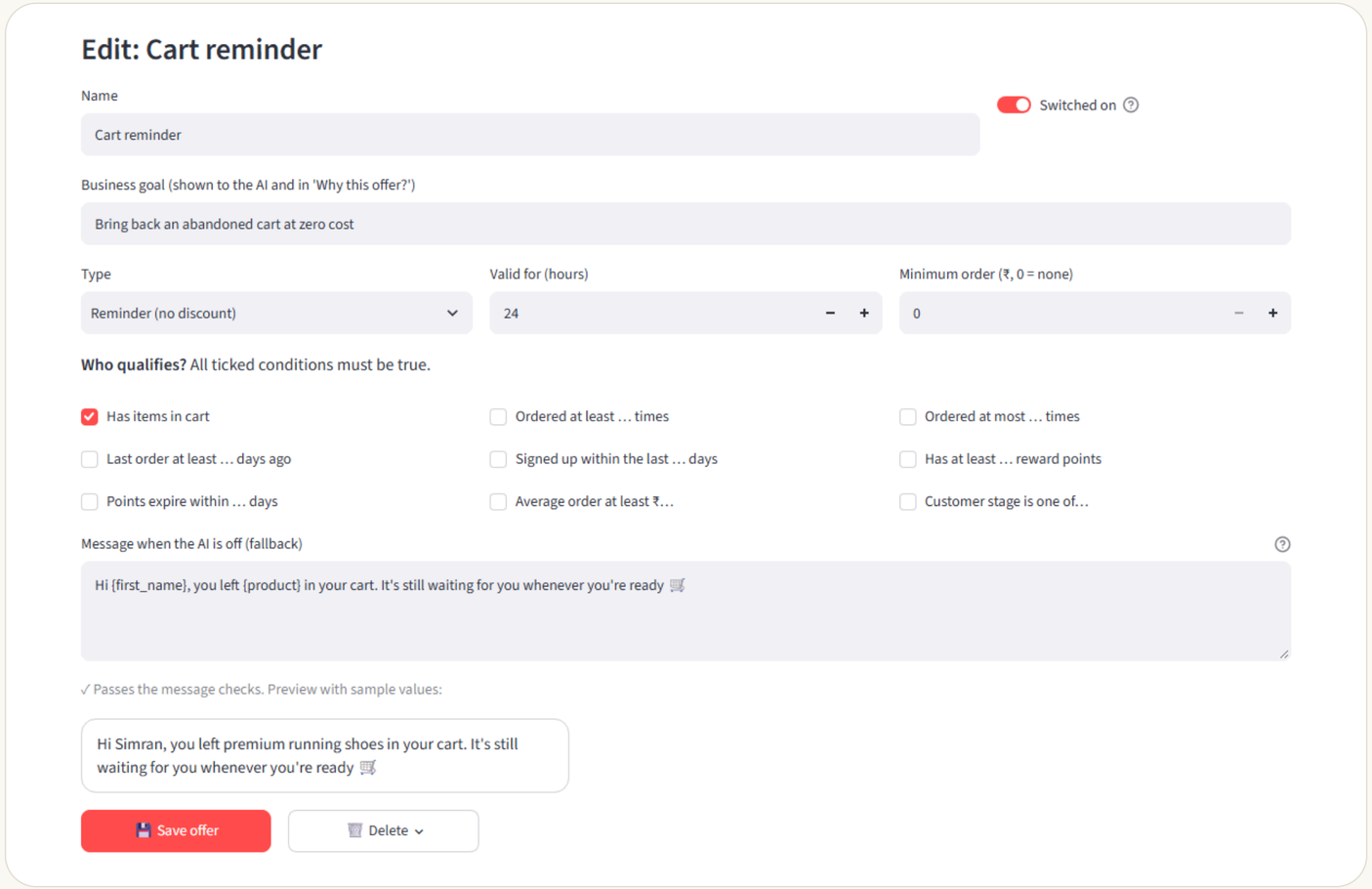


FEATURE 3

Offer manager

Ten offers, from free cart reminders to win-back discounts, in one place.

1. Portal → Offers → choose one
2. Switch it on or off; set its range and minimum order
3. Tick who qualifies: cart, orders, stage…
4. Write the fallback message: checked live, with a preview
5. Save offer: the bot uses it next



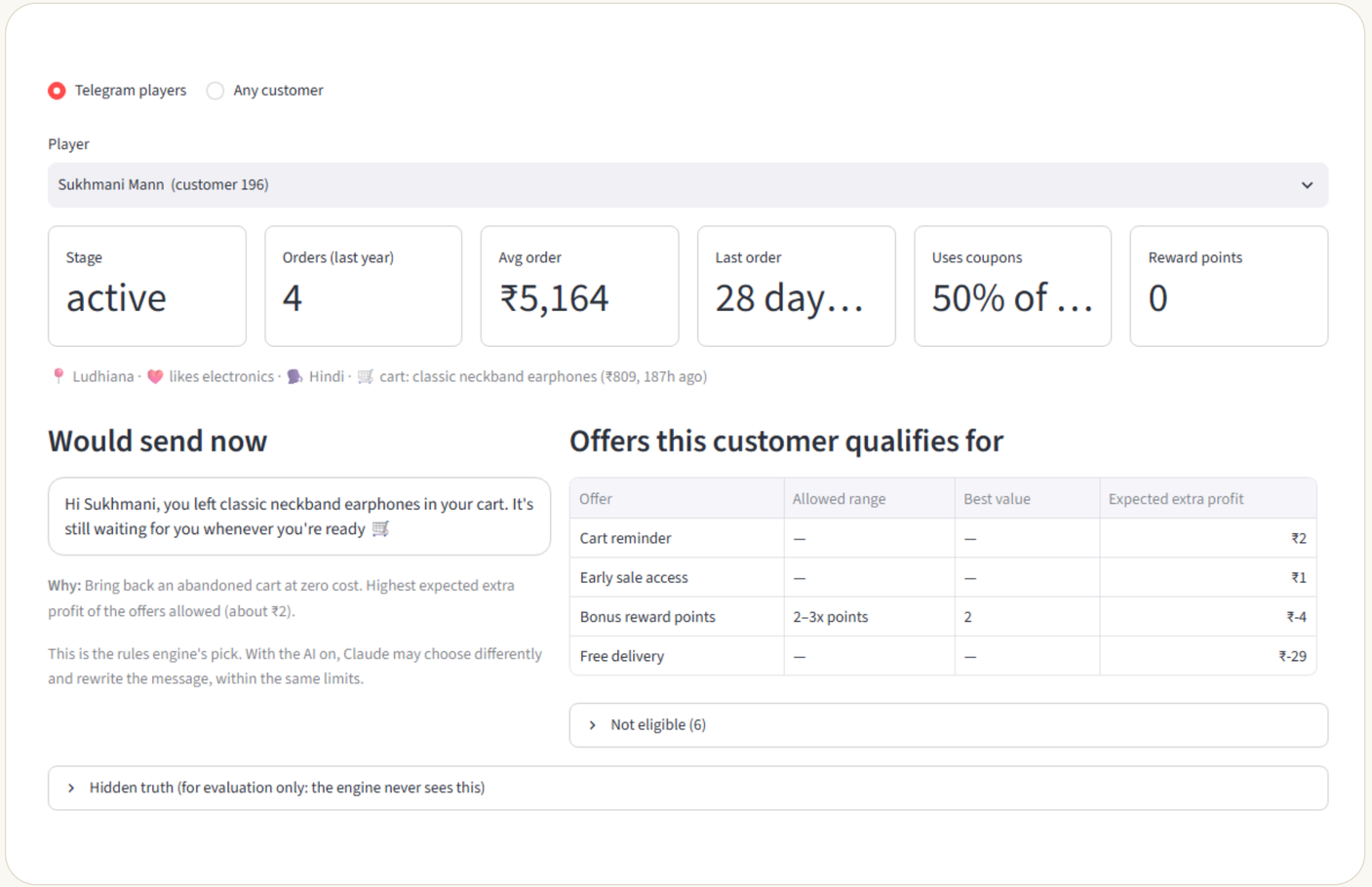
FEATURE 4

Customer preview: why this offer?

See any customer the way the engine does. Nothing is sent.

1. Portal → Customer preview
2. Pick a Telegram player or any customer
3. See the offers they qualify for, with expected extra profit
4. See the exact message they'd get

A discount for someone who'd buy anyway loses money, so the engine may send nothing.



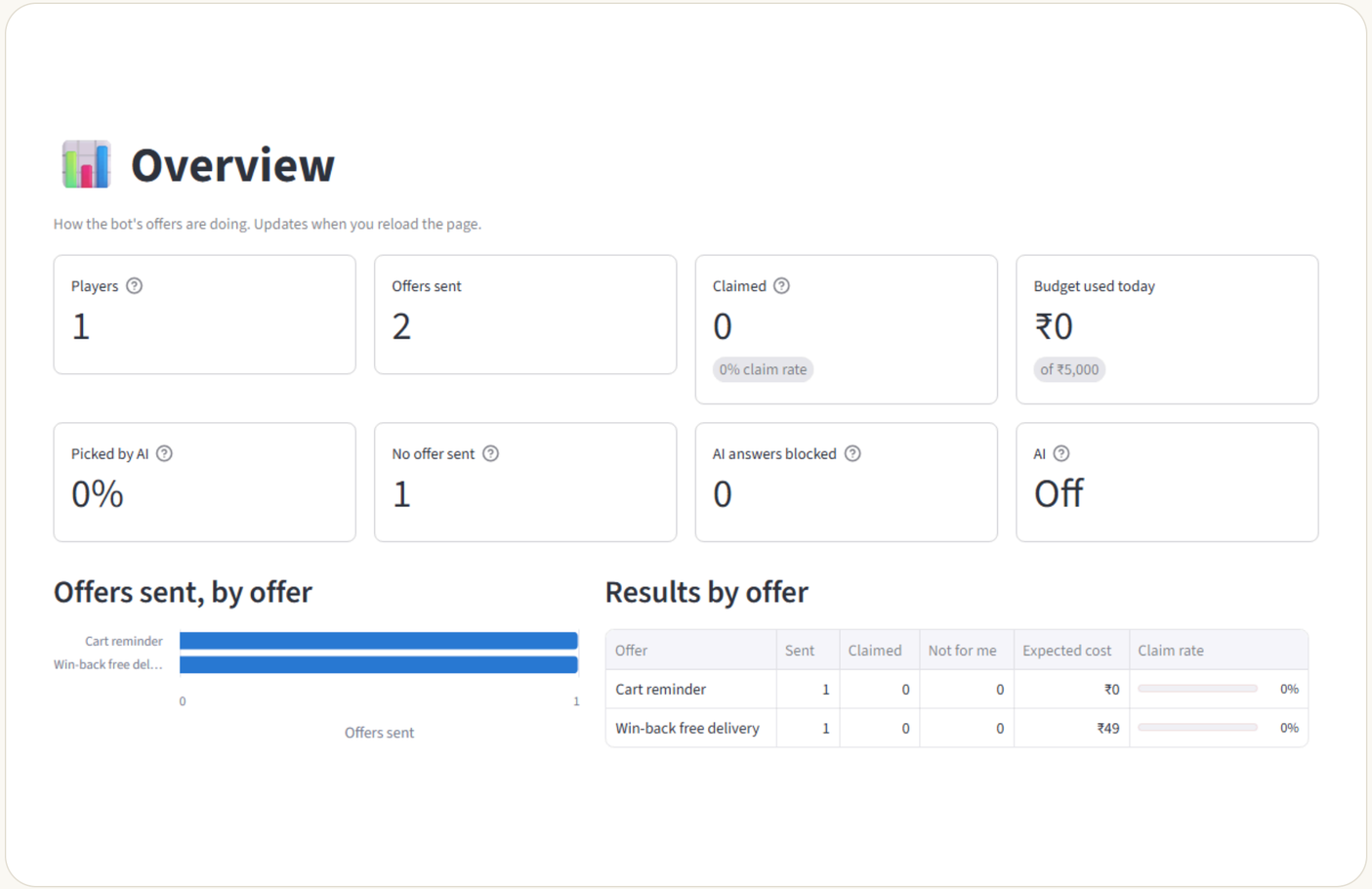
FEATURE 5

Results dashboard

How the offers are doing, at a glance.

1. Portal → Overview
2. Players, offers sent, claim rate, budget used today
3. Share picked by the AI, and AI answers the rules blocked
4. Results per offer, and the latest decisions with reasons

Every decision is logged: what was chosen, by whom, and why.



DATA REALISM

Synthetic shoppers that behave like real ones

66%

of revenue from the top 20% of customers

73% vs 20%

orders with a coupon: deal hunters vs loyal customers

26% vs 8%

returned: fashion vs electronics

1.31×

orders per day at weekends; paydays and festivals spike too

46% vs 13%

cash on delivery: small towns vs Delhi

33%

of customers have an abandoned cart

Each customer has a hidden type the engine never sees; it must infer behaviour from orders. Tests check these ranges.

Links for evaluators

What	Where
Live bot	t.me/oooHusanBot
Marketing portal	python -m streamlit run portal/app.py
Code (public)	github.com/Hussnn18/h2
Testing data	data/sample/*.csv in the repo
Docs	README.md · plan.md · CLAUDE.md

Built with Claude Code · Python · python-telegram-bot · Claude API · SQLite · Streamlit